SANJIVANI UNIVERSITY

DEPARTMENT OF AIDS

UNIT – 1 MINI PROJECT REPORT

on

Enterprise Student Record and Academic Management Platform

Design Thinking and Product Planning

Submitted By :

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Subject: DevOps and Cloud Foundation (IBM)

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CERTIFICATE

This is to certify that **Madhuri Jape** has completed the Unit–1 Mini Project titled "EduBridge – Academic Operations & Student Success Platform" as part of the DevOps and Cloud (IBM) course curriculum.

The report demonstrates an independent application of the prescribed product-planning activities. It covers business understanding, stakeholder analysis, user personas, Design Thinking, IBM Design Thinking, requirements engineering, product vision, MVP planning, roadmap development, initial architecture, project charter and repository planning.

The work is prepared for academic evaluation and represents the planning stage of a software product before detailed Agile implementation.

DECLARATION

I, **Madhuri Shivaji Jape**, hereby declare that this Unit–1 Mini Project Report titled **"EduBridge – Academic Operations & Student Success Platform"** has been prepared for academic submission in the **DevOps and Cloud Foundation (IBM)** course.

The report has been developed according to the activities and deliverables specified in the Unit–1 Mini Project Guide. The reference material was used to understand the required structure, concepts, and learning outcomes, while the project concept, examples, requirements, personas, analysis, and planning decisions have been developed specifically for this submission.

This report represents a **product-discovery and planning exercise** and does not claim that the proposed EduBridge platform has already been fully implemented, deployed, or tested in a production college environment. The work has been prepared for academic and educational purposes.

ACKNOWLEDGEMENT

I sincerely express my gratitude to my subject faculty **Prof. Karan Gaikwad** for providing valuable guidance and support throughout the preparation of this Unit–1 Mini Project Report. His guidance helped me understand the importance of software product engineering, Design Thinking, IBM Design Thinking, requirements analysis, and structured product planning.

I am also thankful to the **Department of AIDS** and the institution for providing the necessary academic environment and resources to undertake this project. I acknowledge the Unit–1 project guide and reference materials provided for the course, which helped me understand the expected activities, deliverables, terminology, and documentation structure.

I am grateful for the opportunity to explore the **EduBridge** concept from a user, business, and product perspective before moving toward detailed software development. The analysis, requirements, personas, and planning presented in this report have been developed as part of my academic learning and project work.

ABSTRACT

EduBridge – Academic Operations & Student Success Platform is a proposed digital platform designed to streamline academic operations and improve visibility into student-related activities. In many educational institutions, academic information and routine processes are distributed across spreadsheets, notices, messaging applications, and separate records. This can result in repeated data entry, delayed communication, fragmented information, and difficulty in tracking pending academic activities.

EduBridge proposes a centralized academic workspace connecting **students, faculty members, department coordinators, and examination staff** through role-specific interfaces. The platform is envisioned to support activities such as course registration, attendance monitoring, assessment schedules, internal marks, academic requests, notifications, and institutional reporting.

The proposed system also includes workflow-oriented features that can help identify pending tasks and provide timely reminders or notifications. This can reduce dependence on manual follow-ups and provide users with better visibility of academic activities.

This Unit–1 mini project focuses primarily on **product discovery, Design Thinking, and product planning** rather than full implementation. The report applies the five stages of Design Thinking—**Empathize, Define, Ideate, Prototype, and Test**—along with the three principles of IBM Design Thinking and the **Observe → Reflect → Make** loop.

The report further presents stakeholder analysis, user personas, business and system requirements, functional and non-functional requirements, product vision, success measures, MVP scope, product roadmap, initial system architecture, project milestones, repository structure, and product discovery documentation. These activities establish a foundation for future Agile development and implementation of the proposed EduBridge platform.

Keywords: Academic Operations, Student Success, Student Records, Workflow Automation, Design Thinking, IBM Design Thinking, DevOps, Requirements Engineering, MVP, Product Roadmap.

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1. INTRODUCTION AND PROJECT OBJECTIVE

Software product engineering begins with understanding the environment in which a system will operate. For an academic institution, the challenge is not simply to store student information. The product must support recurring academic activities, clearly separate responsibilities and make important information available when decisions are required.

1.1 Project Objective

- Study the academic-management workflow from the viewpoint of multiple users.
- Identify friction in registration, attendance, assessment, academic requests and reporting.
- Use Design Thinking to move from observed needs to a testable product concept.
- Apply IBM Design Thinking principles to keep the product focused on measurable user outcomes.
- Define functional and non-functional requirements before implementation.
- Create an MVP that is small enough for an initial Agile release.
- Prepare an architecture and repository structure suitable for later DevOps activities.

1.2 Project Scope

The scope covers product discovery and planning. It includes academic workflow analysis, personas, requirements, prototype planning, MVP definition, roadmap and architecture. Coding, cloud provisioning, CI/CD execution and production monitoring are intentionally treated as later-stage activities.

1.3 Product Concept

EduBridge combines academic records with workflow visibility. A student can see relevant academic information and pending actions; faculty can complete academic tasks; coordinators can monitor exceptions; and examination staff can manage assessment-related records. The concept emphasizes clarity and timely action rather than adding a large number of features.

2. BUSINESS UNDERSTANDING

The Unit–1 brief requires the student to understand admissions, course registration, attendance, examinations and result management before defining the software. EduBridge therefore treats these operations as connected stages in one academic journey.

Academic Activity	Typical Input	Operational Difficulty	EduBridge Opportunity
Student onboarding	Student and program details	Repeated entry across forms	Unified student profile
Course registration	Semester subjects and choices	Approval status is hard to track	Registration workflow with status
Attendance	Class-wise attendance	Students discover shortages late	Attendance summary and threshold notice
Assessment	Schedules, marks, components	Information scattered across notices	Assessment calendar and marks view
Results	Validated semester outcomes	Report preparation can be repetitive	Structured result publication
Academic requests	Bonafide, correction, approvals	Manual follow-up and queues	Request tracking with ownership
Reporting	Department-level summaries	Data assembled from multiple files	Role-based reporting workspace

2.1 Business Gaps Identified

- Information may exist but is not always visible to the person who must act on it.
- Different academic activities can maintain separate versions of similar student data.
- Students may depend on notices or manual follow-up to understand pending actions.
- Faculty and coordinators may spend time preparing status summaries instead of resolving exceptions.
- Administrative reporting can become repetitive when data is distributed across files.

2.2 Proposed Business Response

EduBridge proposes a role-based academic workspace with a common student record, task status, academic calendar, controlled updates and report views. The platform is planned to reduce unnecessary handoffs while keeping responsibilities explicit.

3. BUSINESS PROBLEM AND REQUIREMENTS

3.1 Core Problem Statement

Higher-education academic activities are often handled through disconnected workflows. Students need to know what is completed and what requires action, faculty need efficient ways to update academic information, and administrators need reliable summaries. When these needs are supported by separate records and manual communication, delays and duplicated effort can occur.

3.2 Business Requirements

ID	Business Requirement	Expected Value
BR-01	Maintain one authoritative student academic profile.	Less duplication and clearer records.
BR-02	Track course registration from request to approval.	Students can see current status.
BR-03	Provide attendance summaries and threshold notices.	Earlier awareness of attendance issues.
BR-04	Maintain assessment schedules and marks.	More organized academic information.
BR-05	Support controlled academic requests.	Clear ownership and traceability.
BR-06	Provide role-specific dashboards.	Relevant information without clutter.
BR-07	Generate structured academic reports.	Reduce repetitive compilation.
BR-08	Maintain an audit trail for important changes.	Improved accountability.
BR-09	Support future integration and scaling.	Allows later product growth.

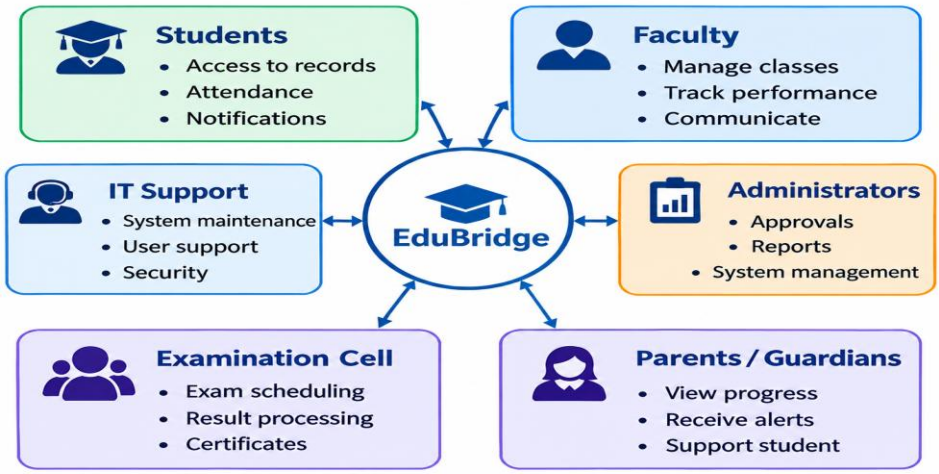
3.3 Business Outcome

The intended outcome is a product plan that can move into Agile development with a clear understanding of users, workflows, priorities and acceptance expectations. The platform is not intended to replace every institutional system in its first release; it is intended to prove a coherent academic workflow experience.

4. STAKEHOLDER ANALYSIS

Stakeholder analysis identifies who contributes information, who uses the product, who approves actions and who maintains the technical environment.

Stakeholder	Interest	Main Responsibility	Expected Product Support
Students	High	Use academic services and monitor status	Simple personal dashboard and alerts
Faculty	High	Record attendance/marks and review requests	Fast entry and action queues
Department Coordinator	High	Monitor course and student-level issues	Department dashboard and exceptions
Examination Cell	High	Manage schedules and result processes	Controlled exam/result workspace
Academic Office	Medium-High	Validate institutional records	Search, reports and audit trail
IT Support	Medium	Maintain access, infrastructure and support	Administration, logs and diagnostics
Management	Medium	Review institutional trends	Summary reports and indicators



4.1 Responsibility Matrix

Activity	Student	Faculty	Coordinator	Exam Cell	IT
Course registration	R	C	A	-	I
Attendance	I	R	A	-	I
Assessment entry	I	R	C	A	I
Result publication	I	C	A	R	I
Access management	I	C	A	-	R
System maintenance	I	I	C	I	R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

5. USER PERSONA DEVELOPMENT

Personas convert stakeholder categories into realistic user goals, behaviors and frustrations. They are used to prevent the product from being designed only around technical functions.

Persona	Goal	Challenges	Expected Experience
Neha – Student	Understand academic status and complete tasks on time.	Multiple notices, unclear pending actions.	One personal view of status and next steps.
Prof. Kulkarni – Faculty	Update academic data without repetitive work.	Time pressure and scattered lists.	Fast entry with clear pending items.
Mr. Deshmukh – Coordinator	Spot unresolved academic issues.	Department data is difficult to summarize.	Exception-focused dashboard.
Ms. Shah – Exam Officer	Process exam information accurately.	Multiple sheets and validation steps.	Controlled exam/result workflow.
Riya – IT Support	Keep access and services dependable.	User access issues and support requests.	Admin tools, logs and clear role controls.

5.1 Persona Insights

- Students need clarity more than additional complexity.
- Faculty workflows should minimize clicks during time-sensitive academic work.
- Coordinators need exception visibility instead of reading every record.
- Examination staff require controlled processes because result data is sensitive.
- IT support needs maintainability, role management and operational visibility.

5.2 Design Implication

The product should present information according to the user's task. A student should not see administrative controls; an examination officer should not have to navigate a student-style dashboard. This supports role-based access and task-oriented interface design.



6. DESIGN THINKING

Design Thinking is a human-centered approach to problem solving that focuses on understanding users, identifying their real problems, generating ideas, creating solutions, and testing them with users.

For the **Enterprise Student Record and Academic Management Platform**, Design Thinking helps us understand the needs of **students, teachers, administrative staff, and IT support staff** before developing the actual software.

6.1 Stages of Design Thinking

Stage	Purpose	Activities
1. Empathize	Understand users and their problems	Observation, interviews, workflow study
2. Define	Clearly identify the main problem	Problem statements, user needs, pain points
3. Ideate	Generate possible solutions	Brainstorming, idea generation
4. Prototype	Create a basic version of the solution	Wireframes, screen designs, mock-ups
5. Test	Check the solution with users	User testing, feedback, improvements

6.2 Design Thinking Cycle

Empathize → Define → Ideate → Prototype → Test → Improve

It is not always a straight-line process. Feedback from testing can send the team back to **Ideate, Define, or even Empathize**.



7. DESIGN THINKING – STAGE 1: EMPATHIZE

The Empathize stage focuses on understanding the people behind the workflow. For this academic project, the discovery plan considers observation, short interviews, workflow walkthroughs and review of existing forms or registers.

7.1 Discovery Questions

Area	Questions
Registration	Where does a student begin, and how does the student know registration is approved?
Attendance	When does a student first realize that attendance is becoming a concern?
Assessments	How are schedule changes communicated and confirmed?
Marks	Who enters marks, who verifies them and who can correct them?
Requests	How does a student know who currently owns a request?
Reporting	Which summaries are prepared repeatedly by staff?
Support	What problems require IT intervention most often?

7.2 Pain-Point Themes

- Status information is often distributed between people, messages and documents.
- Users may know that an action exists but not who is responsible for the next step.
- Academic information becomes more difficult to interpret when it is separated by process.
- Manual reporting can consume time that could otherwise be used for academic support.
- Different users need different levels of detail from the same underlying record.

7.3 Empathy Outcome

The discovery stage suggests that the product should make status, ownership and next actions visible. The goal is not merely digitization; it is to reduce uncertainty in everyday academic work.

8. DESIGN THINKING – STAGE 2: DEFINE

8.1 Defined Problem

Academic stakeholders need a shared but role-controlled workspace that shows the current status of registration, attendance, assessments, results and academic requests, because fragmented processes make it difficult to know what is complete, what is pending and who must act next.

8.2 User Need Statements

User	Need Statement
Student	I need to see my academic status and pending actions in one place so I can respond before deadlines.
Faculty	I need quick academic data-entry and clear pending work so routine administration does not interrupt teaching.
Coordinator	I need to identify exceptions and unresolved cases so I can intervene without manually checking every record.
Exam Officer	I need controlled workflows and validation so exam and result information remains accurate.
IT Support	I need clear roles, logs and maintainable components so operational issues can be diagnosed safely.

8.3 How Might We Questions

- How might we make academic status understandable without exposing unnecessary data?
- How might we reduce repeated entry while preserving accountability?
- How might we surface exceptions without creating notification overload?
- How might we make the product useful on both student and administrative screens?

8.4 Design Goal

Design an academic operations experience that makes the next required action visible, traceable and appropriate to the user's role.

9. DESIGN THINKING – STAGE 3: IDEATE

Ideation deliberately considers multiple solution directions before narrowing the MVP. Ideas are evaluated against user value, feasibility, risk and their contribution to the core workflow.

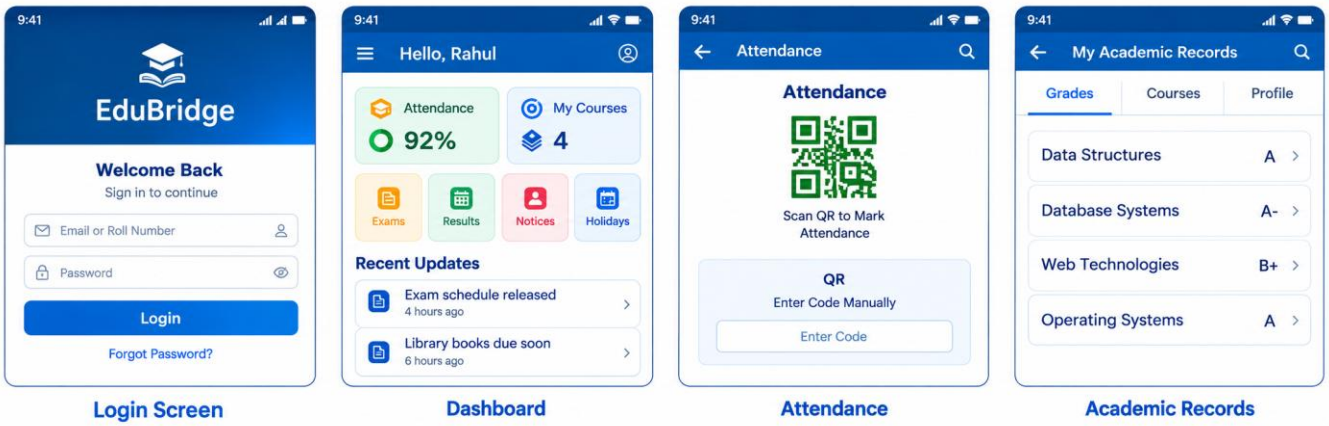
Idea	Purpose	MVP Decision	Reason
Personal academic dashboard	Show relevant status and tasks.	Include	Core student experience.
Registration workflow	Track requests and approvals.	Include	Directly addresses status uncertainty.
Attendance threshold notice	Surface potential attendance concern.	Include	Useful and relatively bounded.
Assessment calendar	Centralize upcoming academic events.	Include	Supports planning.
Academic request tracker	Show owner and status.	Include	Improves transparency.
Smart notification rules	Trigger selected reminders.	Limited	Avoid notification overload in first release.
Advanced analytics	Explore trends across cohorts.	Later	Requires stable data and metrics.
AI academic assistant	Answer student questions.	Later	Needs validation after core workflows.

9.1 Idea Prioritization Logic

Ideas that directly reduce uncertainty in the core academic workflow are prioritized for the MVP. Features that require large datasets, complex external integrations or advanced intelligence are intentionally deferred. This keeps the first product testable and prevents scope from expanding faster than learning.

9.2 Concept Summary

Student/Staff Login → Role Dashboard → Academic Workflow → Status Update → Action/Approval → Report or Alert



10. DESIGN THINKING – STAGE 4: PROTOTYPE

The prototype is a low-fidelity representation of the proposed workflow. Its purpose is to validate navigation and task understanding before implementation effort is invested.

10.1 Proposed Screens

Screen	Key Elements	Validation Focus
Login	Role-aware authentication and recovery	Can users understand how to enter the system?
Student Home	Attendance, registration, assessments, pending actions	Can the next action be found quickly?
Faculty Worklist	Classes, attendance entry, marks, pending approvals	Can routine tasks be completed efficiently?
Coordinator View	Exceptions, course status, unresolved cases	Can staff identify what needs attention?
Exam Workspace	Schedules, marks validation, result publication	Are sensitive actions clear and controlled?
Request Tracker	Request status, owner, history	Can users understand progress?
Reports	Filters, summary metrics, export action	Can users obtain useful summaries?

10.2 Prototype Navigation

LOGIN → ROLE CHECK → DASHBOARD → SELECT TASK → ENTER/VIEW DATA → CONFIRM → STATUS UPDATE

10.3 Prototype Rules

- Use consistent labels for status values.
- Show only actions allowed for the current role.
- Require confirmation for sensitive changes.
- Keep high-frequency tasks near the main dashboard.
- Display the owner and current status of workflow requests.

11. DESIGN THINKING – STAGE 5: TEST

Testing in Unit–1 is planned as prototype validation rather than production performance testing. The objective is to learn whether users can understand and complete representative tasks.

Test Scenario	Participant	Success Signal
Check registration status	Student	Finds status and next step without assistance.
Record attendance	Faculty	Completes task with minimal navigation.
Review an exception	Coordinator	Identifies unresolved item quickly.
Validate result record	Exam Officer	Understands validation and publication states.
Track an academic request	Student/Staff	Can identify owner and current status.
Generate summary	Coordinator	Can filter and interpret report output.

11.1 Feedback Questions

- What part of the screen was unclear?
- Which information did you expect to see first?
- Did you know what action to take next?
- Did any label or status appear ambiguous?
- Was any notification unnecessary or confusing?

11.2 Iteration Plan

Feedback is recorded as observations, grouped into usability issues, and converted into prototype changes. The updated prototype is tested again. This iterative cycle is consistent with the guide's emphasis on learning before expensive implementation.

Design Thinking summary:

Empathize → understand users; Define → frame the problem; Ideate → generate alternatives; Prototype → make the idea visible; Test → learn and improve.

12. IBM DESIGN THINKING APPLICATION

IBM Design Thinking in this project is applied as a practical product-planning approach. The emphasis is on user outcomes, continuous learning and collaboration among people with different responsibilities.

IBM Principle	Application in EduBridge	Evidence/Measure
Focus on User Outcomes	Measure whether users can complete academic tasks with less uncertainty and fewer unnecessary handoffs.	Task completion, pending-action clarity, user feedback.
Restless Reinvention	Treat prototype decisions and workflow rules as changeable when evidence shows a better approach.	Iteration count, feedback-driven changes.
Diverse Empowered Teams	Bring student, faculty, administrative, examination and IT perspectives into discovery.	Stakeholder participation and review notes.

12.1 Outcome Thinking

A technical output might be described as “build an attendance module.” The corresponding user outcome is “allow a faculty member to record attendance accurately while making the student's current attendance status understandable.” The second statement provides a stronger basis for product validation.

12.2 Restless Reinvention

EduBridge is not treated as a fixed specification. Labels, dashboard priorities, notification rules and workflow steps can change after prototype feedback. The objective is to improve the product based on evidence rather than protect the first design.

12.3 Diverse Empowered Teams

The planning team should include domain users and technical roles. Students and faculty explain real workflows; administrative and examination users explain governance; developers and DevOps contributors identify implementation implications; UX contributors keep the experience understandable.

13. IBM LOOP – OBSERVE → REFLECT → MAKE

The IBM Loop provides a compact way to continue learning after the first prototype or release. In EduBridge, the loop is applied to academic workflows and user behavior.

OBSERVE → REFLECT → MAKE

Stage	EduBridge Activity	Example Learning
Observe	Review prototype sessions, workflow completion, support issues and feedback.	Students repeatedly ask where a pending request is located.
Reflect	Group observations and identify the underlying usability or workflow issue.	The request status is technically available but visually buried.
Make	Change dashboard placement, wording or navigation and test again.	Pending requests become a primary dashboard card.

13.1 Continuous Improvement Cycle

User Interaction → Evidence → Reflection → Design Change → Retest → Updated Experience

13.2 Example Loop

Observe: faculty spend time finding incomplete marks entries. Reflect: the system separates incomplete entries from the daily worklist. Make: add an “Incomplete Records” filter and action count. The revised workflow is then tested with faculty users.

13.3 Why the Loop Matters

The loop keeps product planning connected to real usage. It also provides a bridge between Design Thinking and later DevOps practices, where feedback, monitoring and iterative delivery become continuous engineering activities.

14. FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Functional requirements describe what the proposed platform should do. They are intentionally written so that they can later be transformed into user stories and acceptance criteria.

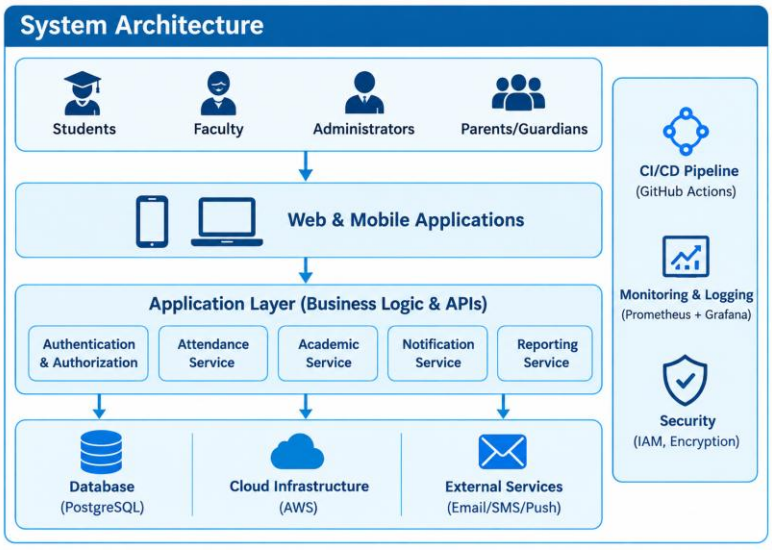
ID	Functional Requirement
FR-01	Authenticate users and provide role-based access for student, faculty, coordinator, examination and support roles.
FR-02	Create and maintain a unified student academic profile.
FR-03	Allow students to view course registration status and approved subjects.
FR-04	Allow authorized faculty to record and update attendance.
FR-05	Calculate and display attendance summaries for authorized users.
FR-06	Maintain assessment schedules and relevant academic events.
FR-07	Allow authorized users to enter, validate and publish marks according to workflow.
FR-08	Allow students to submit defined academic service requests.
FR-09	Track request status, owner, timestamps and history.
FR-10	Generate selected academic summaries for authorized staff.
FR-11	Maintain an audit trail for sensitive updates.
FR-12	Provide dashboard notifications for selected pending actions.

14.1 Functional Flow

Authenticate → Identify Role → View/Enter Data → Validate → Save → Update Status → Notify/Report

14.2 FRS Validation

Each requirement should later receive acceptance criteria and test cases. Requirements that involve permissions should be tested for both allowed and denied access conditions.



15. NON-FUNCTIONAL REQUIREMENTS SPECIFICATION (NFRS)

Non-functional requirements define the quality characteristics expected from the platform. They are important because a feature-rich academic system is not useful if it is insecure, confusing or unreliable.

ID	Category	Requirement
NFR-01	Security	Protect academic records through authentication, authorization and secure transport.
NFR-02	Privacy	Expose student information only to roles with a legitimate operational need.
NFR-03	Usability	Use consistent navigation and understandable status labels across role dashboards.
NFR-04	Performance	Common dashboard and record retrieval operations should respond promptly under expected academic usage.
NFR-05	Reliability	Prevent partial or inconsistent updates during routine transactions.
NFR-06	Data Integrity	Validate required fields and prevent conflicting duplicate records.
NFR-07	Maintainability	Separate business modules and keep documentation current.
NFR-08	Scalability	Allow growth in students, courses, transactions and reports without redesigning the complete product.
NFR-09	Availability	Keep essential academic workflows available during normal operating periods.
NFR-10	Auditability	Record important administrative changes with actor and timestamp information.

15.1 Quality Priorities

- Security and access control are essential because academic information is sensitive.
- Usability is essential because the product serves both technical and non-technical users.
- Data integrity is essential because incorrect academic records can affect downstream decisions.
- Maintainability is essential for future Agile and DevOps development.

16. PRODUCT VISION AND SUCCESS MEASURES

16.1 Vision Statement

To provide a clear and dependable digital workspace through which students and academic teams can manage everyday academic activities, understand current status and complete required actions with fewer manual handoffs.

16.2 Mission Statement

Understand academic workflows, simplify high-frequency tasks, make ownership visible, validate the experience through iterative Design Thinking, and prepare a maintainable product foundation for Agile and DevOps delivery.

16.3 Product Objectives

- Create a consistent academic profile across core workflows.
- Make registration and request status visible to the relevant user.
- Reduce unnecessary manual compilation for routine academic summaries.
- Provide role-specific dashboards rather than one overloaded interface.
- Establish traceability for sensitive academic changes.
- Create a foundation that can evolve through later development units.

16.4 Success Criteria

Measure	What It Indicates
Task completion rate	Whether users can complete key workflows.
Time to locate status	Whether information is easy to find.
Pending-action clarity	Whether users know what to do next.
Error/rework frequency	Whether workflows reduce avoidable mistakes.
User feedback	Whether the experience meets stakeholder expectations.
Requirement coverage	Whether the MVP delivers the planned product promise.

17. MINIMUM VIABLE PRODUCT (MVP)

The MVP is the smallest release that can test the central product idea: a shared academic workspace that makes current status and required actions clear.

MVP Feature	Description	Priority
Secure Login & Roles	Role-specific entry and access control.	Core
Student Profile	Academic identity and basic program information.	Core
Course Registration	View, submit and track registration status.	Core
Attendance Summary	Authorized entry plus student-facing summary.	Core
Assessment Calendar	Upcoming assessment information in one place.	Core
Marks Workflow	Authorized entry, validation and publication states.	Core
Academic Requests	Submit and track selected service requests.	Core
Action Notifications	Selected reminders for pending tasks.	Core
Basic Reports	Role-appropriate academic summaries.	Core
Audit Trail	Record sensitive changes and actors.	Core

17.1 Deferred Features

- Advanced predictive analytics
- Conversational academic assistant
- Large-scale external messaging integrations
- Complex recommendation engines
- Full institutional ERP replacement

17.2 MVP Principle

The MVP should validate the workflow experience before the product expands into advanced intelligence or broad integration. Deferred features remain possible future directions rather than commitments for the first release.

18. PRODUCT ROADMAP

Phase	Focus	Main Output
Phase 1	Domain Discovery	Workflow map, stakeholders, personas and problem definition.
Phase 2	Design Thinking	Ideas, low-fidelity prototype and test plan.
Phase 3	Requirements & Agile Planning	Prioritized backlog, FRS, NFRS and acceptance criteria.
Phase 4	Core Platform	Authentication, profiles, registration and basic academic records.
Phase 5	Workflow Modules	Attendance, assessments, marks and academic requests.
Phase 6	Quality & DevOps	Testing, containerization, CI/CD and environment automation.
Phase 7	Deployment & Improvement	Operational rollout, monitoring and feedback-driven refinement.

19. INITIAL SYSTEM ARCHITECTURE

The architecture is conceptual at Unit–1 stage. It identifies responsibilities without locking the project into a final technology stack.

Users → Web/Mobile Interface → Application/API Layer → Business Modules → Database

19.1 Supporting Components

- Authentication and authorization service for role enforcement.
- Workflow modules for registration, attendance, assessment, requests and results.
- Notification component for selected reminders.
- Audit component for sensitive changes.
- Reporting layer for authorized summaries.
- DevOps layer for automated build, test, deployment and monitoring in later units.

19.2 Architecture Principles

- Keep business modules logically separated.
- Protect data at every access layer.
- Prefer reusable services over duplicated logic.
- Design interfaces so the frontend can evolve independently from core business rules.
- Introduce infrastructure automation after requirements stabilize.

20. PROJECT CHARTER AND MILESTONES

Charter Element	Definition
Project Title	EduBridge – Academic Operations & Student Success Platform
Purpose	Plan a role-based academic workflow product before implementation.
Primary Users	Students, faculty, coordinators, examination staff and IT support.
Scope	Business understanding, Design Thinking, requirements, MVP, roadmap and architecture.
MVP	Profiles, registration, attendance, assessment calendar, marks workflow, requests, notifications and basic reports.
Major Deliverables	BRD, personas, stakeholder analysis, FRS, NFRS, Design Thinking report, IBM documentation, MVP, roadmap, architecture and repository plan.
Constraint	Unit–1 focuses on product discovery and planning rather than full implementation.
Success	A coherent product plan ready for Agile development.

20.1 Milestone Plan

1. Business understanding and problem definition
2. Stakeholder mapping and persona development
3. Design Thinking documentation
4. IBM Design Thinking and Loop documentation
5. Functional and non-functional requirements
6. Product vision and MVP definition
7. Roadmap and architecture concept
8. Project charter and repository structure
9. Final product discovery review

21. REPOSITORY INITIALIZATION AND FOLDER STRUCTURE

```
edubridge-platform/
├── .github/                # CI/CD workflows
├── app/
│   ├── auth/               # authentication and roles
│   ├── students/           # student profiles
│   ├── registration/       # course registration
│   ├── attendance/         # attendance services
│   ├── assessments/        # schedules and marks
│   ├── requests/           # academic requests
│   └── reports/            # reporting services
├── tests/                  # unit and integration tests
├── docs/                   # BRD, FRS, NFRS and design records
├── deployment/             # deployment manifests
├── monitoring/             # monitoring configuration
└── README.md               # setup and project overview
```

22. PRODUCT DISCOVERY DOCUMENTATION

Discovery Area	Decision / Finding
Target Domain	Higher-education academic operations and student services.
Primary User Groups	Students and faculty, with coordinators, exam staff and IT as key operational stakeholders.
Core Problem	Disconnected workflows make status, ownership and next actions difficult to track.
Core Solution	Role-based academic workspace with workflow visibility and controlled records.
Design Approach	Design Thinking plus IBM Design Thinking and Observe → Reflect → Make.
MVP Decision	Prove core academic workflows before advanced analytics or AI features.
Future Direction	Expand reporting, integrations and intelligent assistance after validated usage.

23. CONCLUSION AND FUTURE DIRECTION

EduBridge has been planned as a complete academic operations product rather than a collection of screens. The project focuses on understanding the needs of students, faculty and administrators, applying the Design Thinking approach, and converting the findings into clear requirements, product vision, MVP, roadmap and conceptual architecture. This approach provides a structured understanding of how the proposed system can address common academic management challenges.

The Unit–1 outcome provides a strong foundation for future **Agile and DevOps development**. In the next stage, the identified requirements can be converted into user stories and prioritized for iterative MVP development. The system can then be developed, tested and improved through regular feedback from students, faculty and administrators. DevOps practices can further support automated testing, continuous integration, deployment and system monitoring.

Future enhancements may include **academic analytics, automated notifications, mobile support, improved security, performance monitoring and AI-based academic insights**. Features such as personalized student dashboards, attendance alerts and academic performance analysis can further improve the usefulness of the platform. The **Observe → Reflect → Make** cycle can be continued throughout development to ensure that EduBridge remains user-centered, scalable and adaptable to changing institutional needs.